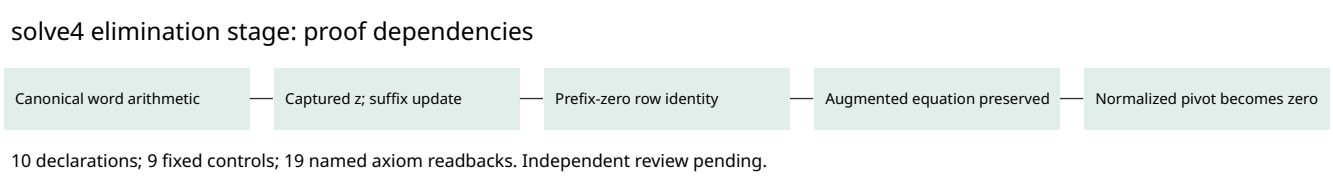
## solve4 single-row elimination: producer evidence

Ten source-linked declarations and nine exact controls passed. The native suffix update j >= col preserves the augmented equation when the pivot row has zero earlier coefficients and satisfies that equation. A normalized pivot makes the target pivot coefficient zero. Separate readback covers all19 declarations; independent source-fidelity review awaits publication.




Source correspondence. Native solve4 captures z before changing any target coefficient and updates columns col through4, including the right-hand side. The word model uses exactly (a+p-b)%p and a*b%p. Canonical inputs, prime p and p <= 2^32 ensure the multiplication and addition intermediates do not overflow UInt64. Outputs remain canonical. Field transvection/equation preservation is standard; this packet derives its correspondence to this specific suffix update.

Fixed controls cover canonical pivot/target rows, the complete expected output row, preserved equation, zero multiplier, failure without prefix zeros, failure when RHS is omitted, failure when the multiplier is recaptured after zeroing the pivot, and failure without normalization. The last four are exact countermodels of omitted obligations. The captured-multiplier control compares the resulting row for this fixture; it does not execute a separate incorrect loop implementation.

Evidence. Nine bounded software invocations: field startup failure, field pass, missing prime-instance word failure, word pass, two control elaboration failures, controls pass, axiom readback pass, and expected false-RHS rejection. All pre/post recursive inventories were unchanged. The first startup failure used4GiB address space; a prospective infrastructure amendment used8GiB address space,3500MiB Lean heap and16MiB thread stack. Each wrapper has120seconds plus3seconds kill reserve and90second child timeout. A shell permission error before starting Lean was corrected by invoking the wrapper with bash. No scientific trajectories were run.

Scope and next construction. This proves value correspondence for one target-row suffix assignment as a finite function. It assumes the pivot-prefix invariant and, for zeroing, normalization to1. Deriving those premises from pivot search, row swap, modular inversion and prior columns is next; actual nested array mutation, four-column induction, operation counters and solve4 return correctness are not yet proved. No performance measurements are reported.

Replay. Source snapshots and check wrapper are committed; the official proof-attempt archive retains exact module versions, compiled outputs, stdout/stderr, recursive hashes and receipts for every invocation. The wrapper uses the recorded local dependency roots. A replay on another host must restore matching source/toolchain/cache identities before checking. Independent review is a separate record after immutable producer publication.